Copenhagen doesn't work.) How have physicists failed to get the memo from philosophers after all this time? Part of the problem is that physicists generally don't know much about philosophy. There is a massive asymmetry between the two fields: while philosophers usually take physics very seriously indeed—philosophers of physics are mathematically conversant in physics and often have advanced degrees in both fields—physicists are rarely trained in philosophy at all. Despite their ignorance of philosophy (or more likely because of their ignorance), some physicists are openly contemptuous of the subject. "Philosophy is dead," declared Stephen Hawking in 2011. "Philosophers have not kept up with modern developments in science. Particularly physics." And according to Neil deGrasse Tyson, studying philosophy "can really mess you up." "Pretty much after quantum mechanics… philosophy has basically parted ways from the frontier of the physical sciences," Tyson claimed. "I'm disappointed because there is a lot of brainpower there, that might have otherwise contributed mightily, but today simply does not." Physicist Lawrence Krauss surmised that this antagonism between physics and philosophy comes from envy on the part of the philosophers, "because science progresses and philosophy doesn't," he claimed. "Philosophy is a field that, unfortunately, reminds me of that old Woody Allen joke, 'those that can't do, teach, and those that can't teach, teach gym.' And the worst part of philosophy is the philosophy of science.… It's really hard to understand what justifies it."

These are breathtakingly ignorant claims. Yet Hawking, Tyson, and Krauss are certainly not stupid people—why do they know so little about philosophy? Their attitudes are even stranger when put in historical perspective. Just a few generations ago, at the birth of quantum physics, all physicists received some schooling in philosophy. Einstein read Mach, Bohr read Kant. But the shifts in research funding and physics classrooms after World War II also led to broader shifts in university curricula. For Einstein and Bohr's generation, philosophy was part of the core educational curriculum in central Europe. But in postwar America, it was (and is) relatively easy for an intelligent student to go all the way from kindergarten to a PhD in physics at a top-tier university without ever darkening the door of a philosophy classroom.

This is not a plea for the good old days. This problem isn't especially